

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 01

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1** The curve $y = x^2 + kx$ (with $k < 0$) crosses the x -axis at the origin and at one other point P . The gradient of the curve at P is 6. Find k .

- A** 6
- B** -6
- C** -12
- D** -3
- E** 3
- F** -2

- 2 Express $\frac{5}{3 + 2\sqrt{5}}$ in the form $a + b\sqrt{5}$ where a and b are rational.
- A** $15 - 10\sqrt{5}$
- B** $\frac{-15 + 10\sqrt{5}}{11}$
- C** $\frac{15 + 10\sqrt{5}}{29}$
- D** $\frac{15 - 10\sqrt{5}}{11}$
- E** $-15 + 10\sqrt{5}$
- F** $\frac{-15 + 10\sqrt{5}}{29}$

3 Let $f(x) = x^2 - 2x - 8$. For how many integers n is $f(n) < 0$?

A 8

B 6

C 5

D 7

E 3

F 4

- 4 Let P : 'the quadrilateral $ABCD$ is a square' and Q : 'the quadrilateral $ABCD$ has four equal sides'. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

- 5 Which of the following statements is/are true?
- I** Doubling the radius of a circle doubles its circumference.
 - II** Doubling the radius of a sphere multiplies its volume by 8.
 - III** Doubling the radius of a cylinder while halving its height leaves its volume unchanged.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 6** Consider the statement S : **for all real x , $x^2 - 6x + 10 > 0$.** Which of the following is true?
- A** S is false, because $x = 3$ gives $x^2 - 6x + 10 = 0$.
 - B** S cannot be decided without knowing x .
 - C** S is false, because a quadratic can be negative.
 - D** S is true, because $x^2 - 6x + 10 = (x - 3)^2 + 1$.
 - E** S is true, because $x^2 - 6x + 10$ has two real roots.

- 7 Consider the claim: **if n is a positive integer then $n^2 + n + 1$ is odd.** Which of the following is a valid proof?
- A** $n^2 + n = n(n + 1)$ is even, being the product of consecutive integers, so $n^2 + n + 1$ is even plus one, which is odd.
- B** $n^2 + n + 1$ cannot be factorised, so it is odd.
- C** For $n = 1, 2, 3$ the values are 3, 7, 13, all odd, so the claim holds.
- D** Suppose $n^2 + n + 1$ is even. Then n is even, which is a contradiction.
- E** If n is odd then n^2 is odd and n is odd, so $n^2 + n + 1$ is odd.

- 8** Which of the following statements about a scatter graph showing strong positive correlation between hours of revision and exam mark is/are valid?
- I** Students who revised more tended to score higher.
 - II** Revising more causes a higher mark.
 - III** A line of best fit could be used to estimate the mark of a student who revised for a number of hours within the range of the data.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 9** A student wants to prove that **the square of any odd integer leaves remainder 1 when divided by 8**.
- Which of the following is the most appropriate first step?
- A** Check the claim for 1, 3, 5, 7 and 9.
 - B** Assume the square leaves a remainder other than 1.
 - C** Assume the claim is true for all odd integers.
 - D** Write the odd integer as $2k$ for some integer k .
 - E** Write the odd integer as $2k + 1$ for some integer k .

10 A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

I Complete the square: $x^2 + 2x + 3 = (x + 1)^2 + 2$.

II $(x + 1)^2 \geq 0$ for all real x .

III So $(x + 1)^2 + 2 \geq 2 > 0$ for all real x .

IV Hence the equation has no real solutions.

Which of the following is correct?

A The proof is completely correct.

B The first error is on line I.

C The first error is on line II: $(x + 1)^2$ can be negative.

D The first error is on line III.

E The first error is on line IV: a positive expression can still equal zero.

11 $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.3$. Which of the following statements is/are true?

- I** A and B are independent.
- II** A and B are mutually exclusive.
- III** $P(A \text{ or } B) = 0.8$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 12** Let $f(x) = \sqrt{x-3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?
- I** $g(f(x)) = x$ for all $x \geq 3$
 - II** $f(g(x)) = x$ for all real x
 - III** $f(g(x)) = |x|$ for all real x
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

13 Claim: $\int_{-1}^1 \frac{1}{x^2} dx = -2$.

I An antiderivative of x^{-2} is $-x^{-1}$.

II So the integral equals $\left[-\frac{1}{x}\right]_{-1}^1 = -1 - 1 = -2$.

III Hence the value is -2 .

Which of the following is correct?

- A** The first error is on line II: the integrand is undefined at $x = 0$, which lies inside the interval, so the fundamental theorem cannot be applied.
- B** The proof is completely correct.
- C** The first error is on line I.
- D** The first error is on line II: $-1 - 1 = 0$.
- E** The first error is on line III: the value should be $+2$.

14 A student writes the following argument.

I $0.1 > 0.01$.

II Taking logarithms to base 10 of both sides: $-1 > -2$.

III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

A All three lines are correct.

B Line II is wrong: taking logarithms reverses an inequality.

C Line III is wrong: $\log_{10} \frac{1}{2}$ is positive.

D Line II is wrong: $\log_{10} 0.1 = 1$.

E Line I is wrong.

15 Which of the following statements is/are true?

I $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$

II $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$

III $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

16 Which of the following statements is/are true about real numbers?

I For all x and all y , if $x < y$ then $x^2 < y^2$.

II For all x there exists y such that $x < y$ and $x^2 < y^2$.

III There exist x and y with $x < y$ and $x^2 > y^2$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 17 Which of the following statements about the graph of $y = \cos x$ is/are true?
- I It is symmetric about the y -axis.
 - II It is a translation of the graph of $y = \sin x$ by $\frac{\pi}{2}$ in the negative x -direction.
 - III It has period π .
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

18 Which of the following is a counterexample to the claim **'for all positive x and y , $\log_{10}(x+y) \leq \log_{10} x + \log_{10} y$ '**?

A $x = 10, y = 100$

B $x = y = 2$

C $x = 5, y = 5$

D $x = 4, y = 4$

E $x = y = 1$

- 19** Consider the claim: ****if x and y are real numbers with $x + y > 10$, then $x > 5$ or $y > 5$.**** Which of the following proves it?
- A** Suppose $x \leq 5$ and $y \leq 5$. Then $x + y \leq 10$, contradicting $x + y > 10$.
 - B** Suppose $x > 5$. Then $y > 10 - x$, so $y > 5$.
 - C** If $x + y > 10$ then the average of x and y is more than 5, so both exceed 5.
 - D** Take $x = 6$, $y = 6$: then $x + y = 12 > 10$ and $x > 5$.
 - E** If $x > 5$ then the conclusion holds; if $y > 5$ then the conclusion holds.

20 a, b and c are real numbers with $a < b < c$. Which of the following statements must be true?

I $a + b < 2c$

II $ab < bc$

III $\frac{a+c}{2} < c$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

END OF TEST